

PRISM: Pareto-optimal Risk-constrained driving Style Manifold

A Distributional Multi-Objective Framework for Safe and Personalized Autonomous Driving

Anonymous Author(s)
Anonymous Institution

Abstract—Existing reinforcement learning (RL) planners for autonomous driving optimize a single scalar reward that encodes one fixed driving style and treats safety as a tradeable objective. This fails to serve users with diverse preferences and provides no formal safety guarantee against catastrophic outcomes. We propose PRISM, a framework that learns a *Distributional Pareto-Optimal* set of driving style policies – covering comfort-seeking, progress-oriented, disciplined, and cautious behaviors – while enforcing a hard Conditional Value-at-Risk (CVaR) safety constraint that is never traded away regardless of style preference. Building on Distributional Pareto-Optimal Multi-Objective Reinforcement Learning (DPMORL), we optimize over the entire return distribution of each style objective rather than expectations alone, producing policies that are not only diverse in style but also robust to return uncertainty. Safety is enforced via a principled two-tier cost function that separates outcome events from leading indicators, with indicator weights derived from their empirical causal lead times to catastrophic outcomes. We evaluate on the nuPlan closed-loop benchmark, the most realistic and demanding publicly available planning benchmark for autonomous driving. Our results demonstrate that PRISM produces meaningfully distinct driving behaviors across the Pareto front, all satisfying the CVaR safety constraint, significantly outperforming scalar-reward baselines on behavioral diversity while maintaining competitive safety performance.

Index Terms—autonomous driving, multi-objective reinforcement learning, safe reinforcement learning, CVaR, distributional reinforcement learning, driving style personalization, nuPlan

I. INTRODUCTION

Reinforcement learning has emerged as a promising paradigm for autonomous vehicle (AV) planning, with recent methods demonstrating competitive performance on closed-loop benchmarks [1], [2]. However, contemporary RL-based planners share a fundamental limitation: they define a single scalar reward function that encodes one fixed driving style and optimizes for it exclusively. This design choice has two critical consequences.

Problem 1: Single driving style. The weights of a scalar reward encode a fixed behavioral trade-off. Different users have genuinely different driving preferences – some prefer smooth, comfort-oriented riding; others prefer assertive, progress-seeking behavior [3], [4]. A single policy cannot serve all users, and retraining for each user is impractical. This inflexibility is recognized as a significant barrier to user acceptance and trust in autonomous vehicles [5].

Problem 2: Expectation-only optimization and safety as a tradeable objective. Current multi-objective RL (MORL)

approaches for driving optimize only the *expected* return of each objective [6]. Two policies with identical expected comfort but different variance profiles are treated as equivalent – a policy that is consistently comfortable is indistinguishable from one that is usually comfortable but occasionally very jerky. Furthermore, existing approaches either include safety as just another objective in the Pareto front [6] – allowing safety to be traded away – or use a fixed penalty that distorts the behavioral objectives.

We address both problems simultaneously with PRISM. Our framework makes two key design contributions:

- 1) **Distributional Pareto-Optimal style policies.** Rather than optimizing expected returns, we learn a set of policies whose *entire return distributions* are Pareto-optimal under stochastic dominance [7]. This distinguishes policies that are consistently good from those that are merely good on average.
- 2) **Hard CVaR safety constraint.** Safety is enforced as a non-negotiable constraint – specifically, a Conditional Value-at-Risk (CVaR) bound on the tail of the cumulative safety cost distribution. This means that even in the worst α -fraction of scenarios, the expected safety cost remains bounded. No style objective can trade against this constraint.

The clean separation between style (optimized) and safety (constrained) is the structural backbone of PRISM. It allows users to choose their preferred driving style along a diverse Pareto front with the assurance that all available styles are equally and formally safe.

We evaluate on nuPlan [8], the world’s first large-scale closed-loop planning benchmark for autonomous driving, which provides reactive multi-agent simulation across real-world urban scenarios. Our observation space and base implementation build on CaRL [1], the current state-of-the-art RL planner on nuPlan, which we extend with our multi-objective and safety-constrained training procedure.

II. RELATED WORK

A. RL-Based Autonomous Driving Planners

RL-based planners have demonstrated strong performance on closed-loop benchmarks. CaRL [1] achieves state-of-the-art results on nuPlan by using a simple route completion reward with PPO, demonstrating that reward simplicity combined with

scalable training can outperform complex shaped rewards. PDM [2] combines rule-based proposal generation with neural scoring. Our work builds on CaRL’s observation space and base PPO implementation, extending it with a multi-objective distributional framework.

B. Multi-Objective RL for Autonomous Driving

Surmann et al. [6] propose a single-policy MORL approach for personalized driving that conditions on a preference weight vector at runtime. Their method demonstrates that style personalization is achievable without retraining, but it uses linear scalarization of expected returns and has no formal safety guarantee – safety is one component of the reward rather than a constraint. Our work differs fundamentally in three ways: we optimize over entire return distributions rather than expectations; safety is a hard CVaR constraint rather than a weighted objective; and we evaluate on nuPlan rather than CARLA.

Abouelazm et al. [9] introduce the Preordered Multi-Objective MDP (Pr-MOMDP) with a lexicographic safety hierarchy enforced via Quantile Dominance. Like us, they argue that safety should not be tradeable. However, their approach produces a single policy under a fixed hierarchy, rather than a Pareto front of diverse style policies with a shared safety guarantee. They also evaluate on CARLA rather than nuPlan.

C. Distributional RL and Risk-Sensitive Planning

Distributional RL models the entire return distribution rather than its expectation [10], [11]. DPMORL [7] extends this to the multi-objective setting, defining Distributional Pareto-Optimal (DPO) policies via stochastic dominance. Crucially, DPMORL proves that CVaR-optimal policies belong to the set of DPO policies (Theorem 3 in [7]), which provides the theoretical justification for adding a CVaR safety constraint on top of the distributional Pareto front. Our work is the first to instantiate DPMORL in autonomous driving with a domain-specific safety constraint.

D. Safe RL for Autonomous Driving

Safe RL for autonomous driving has been studied extensively [12]. Most approaches use expected cost constraints [13] or penalty-based methods [14]. Hu et al. [15] propose long and short-term constraints for safe RL, demonstrating that purely long-term constraints allow vehicles to get stuck in hazardous states during training. CVaR-based constraints have been studied in the general safe RL literature [16] but have not been applied to MORL autonomous driving. The review by Abouelazm et al. [17] identifies context-aware reward design and tail-risk safety as open problems, which our work directly addresses.

III. PROBLEM FORMULATION

A. Multi-Objective Markov Decision Process

We model the autonomous driving problem as a Multi-Objective MDP:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathbf{R}, c, \gamma, T) \quad (1)$$

where $\mathcal{S}$ is the state space (bird’s-eye-view semantic segmentation following CaRL [1]), $\mathcal{A}$ is the continuous action space of acceleration and steering commands, $\mathcal{P}$ are transition dynamics, $\mathbf{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^4$ is the vector of style reward functions, $c : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ is the scalar safety cost function, γ is the discount factor, and T is the episode horizon.

B. Style Reward Vector

At each timestep t , the agent receives a four-dimensional style reward vector:

$$\mathbf{r}_t = (r_t^{\text{comfort}}, r_t^{\text{progress}}, r_t^{\text{lateral}}, r_t^{\text{spacing}}) \in (0, 1]^4 \quad (2)$$

The four objectives are motivated by empirical studies of human driving style preferences [3], [18], which consistently identify three primary axes of behavioral variation: longitudinal assertiveness, lateral discipline, and spacing preference. We decompose longitudinal behavior into comfort and progress as they capture orthogonal aspects – smoothness of motion versus assertiveness in traffic.

1) *Comfort Objective*: The comfort objective rewards smooth, jerk-free motion, motivated by studies showing that jerk is the primary determinant of perceived ride comfort [19], [20]:

$$r_t^{\text{comfort}} = \exp\left(-\frac{j_{lon}^2(t) + j_{lat}^2(t)}{\sigma_j^2}\right) \quad (3)$$

where $j_{lon}(t)$ and $j_{lat}(t)$ are the longitudinal and lateral jerk magnitudes at timestep t , and σ_j^2 is the empirical variance of jerk computed from baseline CaRL rollouts. The exponential form bounds the reward in $(0, 1]$ and ensures all objectives share a common scale, making preference weights $\mathbf{w}$ meaningful across objectives.

2) *Progress Objective*: The progress objective rewards assertive behavior in traffic, capturing the behavioral dimension identified as “aggressiveness” in the driving style literature [6], [18]:

$$r_t^{\text{progress}} = r_t^{\text{speed}} \cdot r_t^{\text{accel}} \cdot (0.5 + 0.5 \cdot r_t^{\text{lane}}) \quad (4)$$

The three components capture speed assertiveness, acceleration decisiveness, and lane choice respectively. Their multiplicative combination enforces an AND condition – all three must be satisfied simultaneously for high progress reward – while the lane term floor of 0.5 prevents temporary suboptimal lane positions from zeroing out the reward entirely.

Speed component:

$$r_t^{\text{speed}} = \exp\left(-\frac{\max(0, v_{desired}(t) - v_{ego}(t))}{\beta \cdot v_{desired}(t)}\right) \quad (5)$$

The asymmetric formulation penalizes being below $v_{desired}$ but not above it, since exceeding $v_{desired}$ is handled by the safety cost. Normalization by $v_{desired}$ ensures the signal is scale-invariant across road types (residential vs. highway). β is set empirically from baseline rollouts.

The desired speed is *regime-conditioned*, motivated by the Intelligent Driver Model [21], [22]:

$$v_{desired}(t) = \begin{cases} v_{limit}(t) & \text{free flow} \\ v_{lead}(t) & \text{car following} \\ v_{lane_avg}(t) & \text{congested} \end{cases} \quad (6)$$

The regime is determined by the presence and proximity of a lead vehicle and lane-level traffic density, all directly observable from nuPlan's privileged state. This context-aware formulation addresses the open problem of fixed desired-speed rewards identified by Abouelazm et al. [17].

Acceleration component:

$$r_t^{accel} = 1 - \exp\left(-\frac{|a_{ego}(t)|}{\gamma_a}\right) \quad (7)$$

The magnitude $|a_{ego}|$ captures both assertive acceleration and decisive braking as expressions of driving assertiveness [6]. γ_a is set empirically.

Lane choice component:

$$r_t^{lane} = \frac{\text{lane_index}(t)}{N(t) - 1} \quad (8)$$

where lane_index is 0 for the rightmost (slowest) lane and $N-1$ for the leftmost (fastest) lane. When $N = 1$, $r_t^{lane} = 0.5$. Lane choice is treated as a progress signal rather than a lateral discipline signal, consistent with findings that lane preference is a primary behavioral marker of driving assertiveness [17].

3) *Lateral Discipline Objective*: The lateral discipline objective rewards precise lane-keeping, motivated by studies of lateral control behavior [23], [24]:

$$r_t^{lateral} = r_t^{dev} \cdot r_t^{heading} \quad (9)$$

The multiplicative combination enforces simultaneous lane centering AND correct heading – a disciplined driver must satisfy both conditions:

Lane deviation component:

$$r_t^{dev} = \delta_d + (1 - \delta_d) \cdot \exp\left(-\left(\frac{d_{lat}(t)}{\sigma_d}\right)^2\right) \quad (10)$$

where $d_{lat}(t)$ is the lateral distance from the lane centerline, σ_d is the tolerance width (set to 0.2m following empirical lane-keeping data [24]), and $\delta_d > 0$ is a floor value that prevents lane changes from being penalized as absolute violations – consistent with observations that humans do not strictly adhere to the lane center [5].

Heading error component:

$$r_t^{heading} = \exp\left(-\left(\frac{|\Delta\psi(t)|}{\phi}\right)^2\right) \quad (11)$$

where $\Delta\psi(t)$ is the heading angle error relative to the lane centerline direction. Magnitude is used since heading error in either direction is equally undisciplined. ϕ is set empirically.

4) *Spacing Objective*: The spacing objective rewards preferred following-distance behavior, capturing the spacing dimension of driving style identified in [3], [18]:

$$r_t^{spacing} = \delta_s + (1 - \delta_s) \cdot \left(1 - \exp\left(-\left(\frac{\max(0, TTC(t))}{\tau}\right)^2\right)\right) \quad (12)$$

where Time-to-Collision is:

$$TTC(t) = \frac{d_{lead}(t)}{v_{ego}(t) - v_{lead}(t)}, \quad v_{ego}(t) > v_{lead}(t) \quad (13)$$

and $TTC(t) = \infty$ when there is no lead vehicle or ego is not closing. TTC is preferred over time headway (THW) for this objective because it captures the actual collision risk accounting for both gap and closing speed [20]. The floor $\delta_s > 0$ ensures that very low TTC – which occurs during legitimate aggressive driving or lane changes – does not collapse the reward. The threshold τ and floor δ_s are set empirically.

Note that low TTC as a *style preference* (aggressive following) is captured here, while low TTC as a *safety hazard* (below critical threshold) is captured separately in the safety cost c_t .

C. Distributional Pareto Optimality

1) *Why Expected Returns Are Insufficient*: Standard MORL optimizes $J_i(\pi) = \mathbb{E}[G_i^\pi]$ where $G_i^\pi = \sum_t \gamma^t r_t^{(i)}$. Two policies with identical expected returns on each objective are treated equivalently even if one has high return variance. For autonomous driving, a comfort policy that is usually smooth but occasionally very jerky is not equivalent to one that is consistently smooth – the entire return distribution matters.

2) *Multi-Objective Return Distribution*: Given policy π , the multi-objective cumulative return is the random vector:

$$\mathbf{Z}^\pi = \sum_{t=0}^T \gamma^t \mathbf{r}_t \in \mathbb{R}^4 \quad (14)$$

with joint distribution $\mu(\pi)$. Following DPMORL [7], we define stochastic dominance over multivariate distributions:

[Stochastic Dominance [7]] $\mu_1 \succ_{SD} \mu_2$ if and only if $\mu_1 \neq \mu_2$ and $\mathbb{E}_{z \sim \mu_1}[f(z)] \geq \mathbb{E}_{z \sim \mu_2}[f(z)]$ for any non-decreasing function $f: \mathbb{R}^4 \rightarrow \mathbb{R}$.

[Distributional Pareto-Optimal Policy [7]] Policy π is Distributional Pareto-Optimal (DPO) if no other policy π' satisfies $\mu(\pi') \succ_{SD} \mu(\pi)$.

DPO extends standard Pareto optimality: instead of finding policies where no objective's *expected value* can be improved without hurting another, we find policies where no objective's *entire return distribution* can be improved. This is a strictly stronger notion of optimality.

Crucially, DPMORL Theorem 3 [7] proves that CVaR-optimal policies belong to the set of DPO policies. This provides the theoretical justification for combining DPMORL with a CVaR safety constraint – the constraint is compatible with distributional Pareto optimality.

D. Safety Cost Function

Safety is enforced through a scalar cost function $c_t \geq 0$ whose cumulative discounted sum $C^\pi = \sum_t \gamma^t c_t$ is a random variable over trajectories. We design c_t with a principled two-tier structure:

$$c_t = c_t^{\text{outcome}} + \sum_j c_t^{\text{lead},j} \quad (15)$$

1) *Outcome Events*: Outcome events are binary, irreversible violations of hard traffic rules. Following nuPlan’s official evaluation protocol [8], we define seven outcome events with severity-ordered weights:

$$c_t^{\text{outcome}} = \sum_{e \in \mathcal{E}} W_e \cdot \mathbb{1}_e(t) \quad (16)$$

TABLE I
OUTCOME EVENT WEIGHTS

Event	Weight W_e	nuPlan Reference
VRU collision (pedestrian/cyclist)	100	[8]
Vehicle collision	80	[8]
Wrong direction driving	70	[8]
Red light violation	50	[8]
Stop sign violation	50	[8]
Drivable area violation	40	[8]
Object collision	20	[8]

VRU collisions receive the highest weight consistent with nuPlan’s zero-tolerance threshold for pedestrian and cyclist collisions [8]. Wrong direction driving is treated as catastrophic following nuPlan-R [25].

2) *Leading Indicator Cost*: Leading indicators are continuous or binary signals that precede outcome events and provide a dense learning signal during training. Without them, the agent receives almost no gradient signal during the majority of training steps where no outcome event occurs [15], [17].

We define five leading indicators, each grounded in traffic law or nuPlan’s official metrics:

- 1) **TTC below 1.5 seconds** – imminent collision risk [20]
- 2) **THW below 2 seconds** – legal tailgating threshold [26]
- 3) **Speed exceeding road-type threshold** – $v_{ego} > v_{limit}(1 + \delta_{road})$ [8]
- 4) **Lane change with agent in blind spot** – nuPlan lane change min-gap metric [8]
- 5) **Predicted red light violation** – estimated arrival time exceeds green phase [27]

Principled weight derivation. Rather than hand-tuning indicator weights, we derive them from the causal relationship between each indicator and the outcome events it can precede. For indicator j with outcome set $\mathcal{O}_j$:

$$w_j^{\text{lead}} = \sum_{i \in \mathcal{O}_j} \frac{W_i}{|\mathcal{O}_j| \cdot T_{j \rightarrow i}} \quad (17)$$

where $T_{j \rightarrow i}$ is the empirically measured mean lead time in timesteps from indicator j to outcome i , estimated from

baseline CaRL rollouts. This ensures that an indicator firing for exactly its natural lead time produces a cumulative cost equal to the average weight of the outcomes it can cause, preserving the safety hierarchy.

Severity signals. Each indicator has a continuous severity signal bounded in $[0, 1]$:

$$s_{TTC}(t) = \max\left(0, \frac{\tau_{TTC} - TTC(t)}{\tau_{TTC}}\right) \cdot \mathbb{1}[TTC(t) < \tau_{TTC}] \quad (18)$$

$$s_{THW}(t) = \max\left(0, \frac{\tau_{THW} - THW(t)}{\tau_{THW}}\right) \cdot \mathbb{1}[THW(t) < \tau_{THW}] \quad (19)$$

$$s_{speed}(t) = \max\left(0, \frac{v_{ego}(t) - v_{threshold}(t)}{v_{threshold}(t)}\right) \cdot \mathbb{1}[v_{ego}(t) > v_{threshold}(t)] \quad (20)$$

$$s_{blind}(t) = \mathbb{1}[|\dot{y}_{ego}(t)| > \theta_{lc}] \cdot \mathbb{1}[\exists \text{ agent in blind spot at } t] \quad (21)$$

$$s_{redlight}(t) = \mathbb{1}[t_{arrival}(t) > t_{red}(t)] \quad (22)$$

The per-timestep cost for each indicator is:

$$c_t^{\text{lead},j} = w_j^{\text{lead}} \cdot s_j(t) \quad (23)$$

Episode-level cap. For persistent indicators (TTC, THW, speed) whose violation duration can exceed their natural lead time, we apply an episode-level cap to prevent cumulative indicator costs from exceeding outcome event costs:

$$\sum_{t=0}^T c_t^{\text{lead},j} \leq \bar{W}_j^{\text{outcome}} = \frac{\sum_{i \in \mathcal{O}_j} W_i}{|\mathcal{O}_j|} \quad (24)$$

Transient indicators (blind spot, predicted red light) whose natural duration matches their lead time do not require capping.

E. CVaR Safety Constraint

A constraint on expected cumulative cost $\mathbb{E}[C^\pi] \leq \epsilon$ is insufficient for safety-critical deployment. Two policies can have identical expected costs while one has frequent catastrophic failures balanced by many safe episodes. CVaR directly addresses this by bounding the tail of the cost distribution.

The Conditional Value-at-Risk at confidence level α is:

$$\text{CVaR}_\alpha(C^\pi) = \mathbb{E}[C^\pi \mid C^\pi \geq \text{VaR}_\alpha(C^\pi)] \quad (25)$$

This represents the expected cost in the worst $(1 - \alpha)$ fraction of trajectories. We enforce:

$$\text{CVaR}_\alpha(C^\pi) \leq \epsilon \quad (26)$$

At $\alpha = 0.95$: even in the worst 5% of scenarios, the expected safety cost is bounded by ϵ .

F. Full Optimization Problem

PRISM finds a set of Distributional Pareto-Optimal policies $\Pi^* = \{\pi_1, \dots, \pi_K\}$ where each policy π_k solves:

$$\max_{\pi_k} \mathbb{E}_{z \sim \mu(\pi_k)} [f_{\theta_k}(z)] \quad \text{s.t.} \quad \text{CVaR}_\alpha(C^{\pi_k}) \leq \epsilon \quad (27)$$

where $f_{\theta_k} : \mathbb{R}^4 \rightarrow \mathbb{R}$ is a non-linear, non-decreasing utility function over the style return vector, learned as part of the DPMORL procedure. Each policy corresponds to a different distributional preference over style objectives, and all policies satisfy the same safety guarantee.

IV. PRISM TRAINING ALGORITHM

Training proceeds in two stages following DPMORL [7] extended with CVaR-Lagrangian optimization.

A. Stage 1: Utility Function Generation

We generate K diverse non-linear utility functions $f_{\theta_1}, \dots, f_{\theta_K}$ over the style return space $[0, 1]^4$. Each is parameterized by a *non-decreasing neural network* – a network with non-negative weights and non-decreasing activations – ensuring the theoretical requirements for DPO are satisfied [7].

The utility functions are trained to be diverse using the objective from DPMORL [7]:

$$J(\theta_i) = \alpha_{div} \cdot J^{value}(\theta_i) + (1 - \alpha_{div}) \cdot J^{grad}(\theta_i) \quad (28)$$

where:

$$J^{value}(\theta_i) = \min_{j \neq i} \mathbb{E}_{z \sim U([0,1]^4)} [(f_{\theta_i}(z) - f_{\theta_j}(z))^2] \quad (29)$$

$$J^{grad}(\theta_i) = \min_{j \neq i} \mathbb{E}_{z_1, z_2} \left[\left(\frac{\Delta f_{\theta_i}}{\|z_2 - z_1\|} - \frac{\Delta f_{\theta_j}}{\|z_2 - z_1\|} \right)^2 \right] \quad (30)$$

This ensures utility functions differ in both value and gradient, producing genuine coverage of different style preference types.

B. Stage 2: Policy Optimization with CVaR Constraint

For each utility function f_{θ_k} , we train a policy π_k using the following procedure.

1) *State Augmentation*: Following DPMORL Algorithm 1 [7], we augment the state with cumulative style returns:

$$\tilde{s}_t = (s_t, z_t), \quad z_t = \sum_{\tau=0}^t \gamma^\tau \mathbf{r}_\tau \quad (31)$$

updated as $z_{t+1} = z_t + \gamma^t \mathbf{r}_t$. This allows the policy to condition on its own return history, enabling optimization of the full return distribution rather than just expected values.

2) *Scalar Reward via Utility Difference*: The multi-dimensional style reward is transformed into a scalar using the marginal utility:

$$R_t = \gamma^{-t} [f_{\theta_k}(z_{t+1}) - f_{\theta_k}(z_t)] \quad (32)$$

This is the key transformation from DPMORL [7]. The scalar reward is not a fixed linear combination of objectives – it is the incremental change in utility as cumulative returns evolve. Because f_{θ_k} is non-linear, this captures non-linear distributional preferences. DPMORL Theorem 2 proves that the optimal policy under this augmented MDP is identical to the optimal policy under utility function f_{θ_k} [7].

3) *CVaR-Lagrangian Objective*: For each policy π_k , the optimization objective is:

$$\max_{\theta_k} \min_{\lambda_k \geq 0} \mathbb{E} \left[\sum_t R_t \right] - \lambda_k \left(\widehat{\text{CVaR}}_\alpha(C^{\pi_k}) - \epsilon \right) \quad (33)$$

Each policy π_k has its own Lagrange multiplier λ_k because different style preferences produce different trajectory distributions and therefore different safety cost distributions. A shared multiplier would cause the aggressive policy’s constraint violation to tighten the conservative policy’s optimization, distorting the Pareto front.

4) *CVaR Estimation*: At each training iteration, collect N rollouts under π_k . For each rollout i , compute:

$$C^i = \sum_{t=0}^T \gamma^t c_t^i \quad (34)$$

Sort rollouts by C^i in descending order. The CVaR estimate is:

$$\widehat{\text{CVaR}}_\alpha = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} C^i \quad (35)$$

where $\mathcal{T}$ is the set of worst $\lceil (1 - \alpha)N \rceil$ rollouts. At $\alpha = 0.95$ with $N = 512$ rollouts, this averages the top 26 worst-cost trajectories.

5) *Lagrange Multiplier Update*:

$$\lambda_k \leftarrow \max \left(0, \lambda_k + \eta_\lambda \left(\widehat{\text{CVaR}}_\alpha(C^{\pi_k}) - \epsilon \right) \right) \quad (36)$$

When the constraint is violated, λ_k increases, penalizing style optimization; when satisfied, λ_k decreases, allowing the policy to optimize style freely. The constraint self-regulates throughout training.

6) *Base RL Algorithm*: The inner policy optimization uses PPO [28] with Stable Baselines 3 [29], consistent with CaRL’s implementation [1]. The augmented environment $\tilde{\mathcal{M}}$ with scalar reward R_t is a standard single-objective MDP – PPO requires no modification, following DPMORL’s design principle [7].

C. Output: CVaR-Safe Distributional Pareto Front

After training, PRISM produces K policies where:

- Each policy corresponds to a different non-linear utility function over style preferences
- Each policy satisfies $\text{CVaR}_\alpha(C^{\pi_k}) \leq \epsilon$

Algorithm 1 PRISM Training

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1: Stage 1: Generate  $K$  diverse utility functions  
    $f_{\theta_1}, \dots, f_{\theta_K}$   
2: Initialize  $\lambda_k = 0$  for all  $k = 1, \dots, K$   
3: for each policy  $k = 1, \dots, K$  do  
4:   Initialize policy  $\pi_k$  from CaRL checkpoint  
5:   repeat  
6:     Augment state:  $\tilde{s}_t = (s_t, z_t)$   
7:     Collect  $N$  rollouts, compute  $R_t$  via Eq. (32)  
8:     Compute  $C^i = \sum_t \gamma^t c_t^i$  for each rollout  $i$   
9:     Estimate  $\widehat{\text{CVaR}}_\alpha$  via Eq. (35)  
10:    Update  $\pi_k$  with PPO on  $R_t - \lambda_k c_t$   
11:    Update  $\lambda_k$  via Eq. (36)  
12:   until convergence  
13: end for  
14: Output: CVaR-safe Distributional Pareto Front  
    $\{\pi_1, \dots, \pi_K\}$ 
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- No policy’s return distribution is stochastically dominated by another

Together they form a *CVaR-safe Distributional Pareto Front* of driving styles. At deployment, a user selects one policy based on preference; all policies provide equal formal safety guarantees.

V. EXPERIMENTAL SETUP

A. Benchmark: nuPlan

We evaluate on nuPlan [8], the world’s first large-scale closed-loop planning benchmark for autonomous driving. nuPlan provides 1,300+ hours of real-world driving data across four cities (Las Vegas, Boston, Pittsburgh, Singapore) with reactive multi-agent simulation at 10 Hz. Its closed-loop evaluation enforces feedback between the ego planner and background agents, making it significantly more challenging than open-loop benchmarks [25].

We report results on Val14 and Test14-Hard, the standard evaluation splits used by CaRL [1] and PDM [2], enabling direct comparison with existing baselines.

B. Observation Space

Following CaRL [1], we use bird’s-eye-view (BEV) semantic segmentation as input processed by a small CNN. The privileged planning setup provides ground-truth perception – no sensor noise or occlusion – consistent with nuPlan’s planning benchmark protocol.

C. Baselines

- **CaRL** [1]: scalar reward PPO baseline
- **MORL-Linear**: MORL with linear scalarization (no CVaR, no distributional optimization), ablating the DP-MORL component
- **DPMORL-NoCVaR**: full distributional optimization without safety constraint, ablating the CVaR component
- **PRISM (ours)**: full framework

D. Evaluation Metrics

nuPlan Closed-Loop Score (CLS): the official composite metric covering safety violations, comfort, TTC, progress efficiency, and speed limit compliance [8].

Behavioral diversity: we measure the distributional distance between style policies using the Wasserstein distance between their return distributions on the four style objectives. A higher diversity score indicates the Pareto front covers a wider behavioral range.

CVaR safety: we report $\widehat{\text{CVaR}}_{0.95}(C^\pi)$ for each policy and verify constraint satisfaction.

Hypervolume: standard MORL metric measuring the volume of objective space dominated by the Pareto front [7].

VI. NOVELTY ASSESSMENT AND LIMITATIONS

A. Summary of Contributions

PRISM makes four contributions that, in combination, have not appeared in prior work:

- 1) **First application of DPMORL to autonomous driving** with domain-specific style objectives grounded in human preference research.
- 2) **CVaR safety constraint on top of distributional Pareto optimization**, theoretically justified by DP-MORL Theorem 3 [7] and practically motivated by the insufficiency of expectation-based safety.
- 3) **Principled two-tier safety cost function** with indicator weights derived from empirical causal lead times to outcome events, rather than hand-tuned.
- 4) **Context-aware desired speed formulation** for the progress objective, addressing an open problem in reward design identified by [17].

Each individual component has prior art. The combination and its application to nuPlan with formal safety guarantees do not.

B. Limitations

- Comfort and efficiency are combined into a single longitudinal objective; their disentanglement is left for future work.
- Lane choice reward assumes leftmost lane is fastest, which breaks in mandatory turn scenarios (addressed in future work).
- Predicted red light violation relies on nuPlan’s inferred traffic light timing rather than ground truth.
- K discrete policies approximate the continuous Pareto front; continuous preference conditioning [6] is a natural extension.

VII. CONCLUSION

We presented PRISM, a framework for learning a set of safe, diverse autonomous driving policies via Distributional Pareto-Optimal Multi-Objective RL with CVaR safety constraints. PRISM cleanly separates style – which is optimized across a Pareto front of distributional preferences – from safety – which is enforced as a non-negotiable tail-risk constraint. The

result is a set of formally safe policies covering diverse user preferences, validated on nuPlan’s closed-loop benchmark. We believe this work provides a principled foundation for personalized, trustworthy autonomous driving systems.

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